Complete Guide: Installing CUDA, Python, and PyTorch for AI (All GPUs)

1. Introduction

Briefly explain why CUDA, PyTorch, and the correct Python version are important for AI/ML work.
Mention that different GPUs support different CUDA versions, and PyTorch must match that CUDA version.
State that this guide works for NVIDIA RTX, GTX, Tesla, and Quadro GPUs.

2. Step 1 — Check Your GPU Model

Command to run in terminal / PowerShell:

```
nvidia-smi
```

This shows GPU Name, Driver Version, and CUDA Version Supported.

Example output for RTX 4060:

```
NVIDIA-SMI 555.42.02
Driver Version: 555.42.02
CUDA Version: 12.1
```

3. Step 2 — Check CUDA Compatibility

Go to NVIDIA's official CUDA compatibility table:
https://developer.nvidia.com/cuda-gpus
Find your GPU model and note the maximum supported CUDA version.

GPU Model	Max CUDA Version
RTX 4060 / 4070	12.2
RTX 3060 / 3070	12.1
RTX 2080 / 2070	11.8
GTX 1660 / 1650	11.4
Tesla T4	11.8

4. Step 3 — Choose the Correct Python Version

Best versions for AI/ML (stable with PyTorch, TensorFlow, etc.):
Python 3.10 ■ Recommended
Python 3.9 ■ Still safe for older tools
Avoid Python 3.12+ (many AI libs don't fully support it yet)

Download Python: https://www.python.org/downloads/

5. Step 4 — Install PyTorch for Your CUDA Version

Go to the official PyTorch install page: https://pytorch.org/get-started/locally/

Example for CUDA 12.1 (RTX 4060, 3060, etc.):

```
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
```

Example for CUDA 11.8 (RTX 2080, Tesla T4, etc.):

```
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
```

Example for CPU only (no NVIDIA GPU):

```
pip install torch torchvision torchaudio
```

6. Step 5 — Verify Installation

```
import torch
print("PyTorch version:", torch.__version__)
print("CUDA version:", torch.version.cuda)
print("GPU available:", torch.cuda.is_available())
print("GPU name:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "No GPU detected")
```

Expected output (example):

```
PyTorch version: 2.5.1+cu121
CUDA version: 12.1
GPU available: True
GPU name: NVIDIA GeForce RTX 4060
```

7. Step 6 — Test GPU with AI Code

```
import torch

x = torch.rand(5000, 5000).cuda()
y = torch.rand(5000, 5000).cuda()
z = torch.matmul(x, y)
print("Matrix multiplication successful on GPU:", z.is_cuda)
```

8. Troubleshooting Section

CUDA error → Make sure your PyTorch and CUDA versions match.

torch.cuda.is_available() = False → Update GPU driver from: https://www.nvidia.com/Download/index.aspx

ChromeDriver errors (if using Selenium) → Match driver version to Chrome version:
https://chromedriver.chromium.org/downloads

Microphone not recording → Check Windows microphone permissions:

Settings → Privacy & Security → Microphone → Allow apps to use microphone.

9. Useful Links

NVIDIA CUDA GPUs list: https://developer.nvidia.com/cuda-gpus

PyTorch installation page: https://pytorch.org/get-started/locally/

Python downloads: https://www.python.org/downloads/

NVIDIA Driver download: https://www.nvidia.com/Download/index.aspx

10. Recommended Document Format

Use Headings for each step.
Include code blocks for commands.
Include links for every external reference.
Provide example outputs after commands.
Keep it step-by-step so even beginners can follow.